Beijing-Dublin International College

BE (Internet of Things) Degree

Project Handbook (Beijing)

Academic Year 2025/26

EEEN3010J

Jan 2026

Contents

1. Introduction

As part of the Internet-of-Things BE Degree Programme, all Stage 4 students must complete a major project.

The BE project involves each student undertaking a substantial body-of-work in a selected area of Internet-of-Things Engineering. The project runs in Semester 2 and follows a plan drawn up by the student in consultation with their principal supervisor, a member of the academic staff.

The project work may involve elements of research, design, analysis, simulation, construction, measurement and experimentation, with the mix depending on the project. In addition, you will need to conduct a critical literature review and enhance your communication skills through a presentation of your results.

Projects will typically cater for multiple students, with each student normally working independently on agreed aspects of the project and keeping in regular contact with their partner, exchanging ideas and results. The division of the workload on joint projects is usually decided by the students themselves in consultation with their supervisor.

1.1 – Projects are student led

Project work is intended to encourage independent thought and critical analysis on the part of each student. It is important to understand that the project is not organised like most of the laboratories you have completed to date. There are no “step-by-step instructions” or prescribed sequence of activities. Instead, guided by their supervisor, individual students are expected to develop their own ideas for their project and determine how they should be implemented. It is therefore important that you keep in regular contact with your supervisor to ensure that the project is developing along the right lines.

1.2 – Project workload

The final year project module is a 15-credit module and requires a total workload of approximately 300-350 hours per student. The following table provides an approximate indication of the total number of hours that you should expect to spend working on your project:

	Pre Semester	**Semester 2**
Preparation (recommended)	10	0
Practical	0	300-350
Total Workload (semester 2)	**300-350 hrs total (about 21-25 hours per week for 14 weeks)**	

1.3 – How the project is assessed

The overall project grade is derived from several components assessed during semester 2 at the dates specified in 0.

1. **Group thesis and individual interview: 70%**
   At the end of the semester each project group must submit a single **final thesis** documenting all aspects of the project. Soon after each student will individually attend an **oral examination** conducted by your supervisor along with a “second reader”.
   Approximately 20% of the grade for this component (i.e. 14% overall) will be contributed to by the “shared” text in the group thesis – the remainder (56% overall) is awarded for the “individual” text component and subsequent oral examination.

2. **Group presentations: 20%**
   At the end of semester each project group must present their work as a group. Presentations are delivered in front of examiners and students. Individual grading can be applied to members of the group. All group members are expected to participate equally in the preparation and delivery; unequal participation will reduce grade awarded. Both technical content and presentation quality will be judged.

3. **Conduct grade: 10%**
   Throughout the semester the principal supervisor will maintain a discretionary score to reflect each student’s conduct including: participation, engagement, timeliness, ability to follow a plan, demonstration of independent research including additional reading, responsiveness and general attitude.

   To gain a good conduct mark you should first engage with your supervisor and be open about your progress. Confident students who feel that they do not need close supervision should nonetheless be careful to meet their supervisors just as frequently as other students, so that the supervisor can assess your level. Note that a good project does not ensure a good conduct mark – always remain visible to your supervisor, interact well, ask probing questions, and seek out feedback.

Late submissions and plagiarism are taken very seriously and dealt with according to the rules set out in Appendix B and Appendix C, respectively.

2. Day-to-Day running of the project

The 15 credits awarded for the project are earned over one semester in Stage 4; with all 15 credits being assigned to Semester 2. The period between allocation of your project in Semester 1, and the commencement of Semester 2, is a useful time to carry out some background reading, in consultation with your supervisor.

2.1 – Beginning the project

After the allocation of projects, you are expected to contact your supervisor as soon as practicable to discuss details of your project. The responsibility is on you, the student, to make initial contact with your supervisor. Since your supervisor will likely not be based in Beijing at that time, you should contact them via email. This initial contact will be followed by a period where additional information relevant to the project background and objectives is collected from different sources e.g. library journals, data sheets, internet sources etc. This *literature review* is an important part of reading yourself into the project and understanding what previous work has been done in the field.

2.2 – Hazard identification / risk analysis

An essential part of the preliminary work is to assess how the project can be conducted safely. You should review the safety aspects of the project and **if necessary** you should prepare a document identifying potential hazards and how they will be managed. You need to consider:

1. Hazard Identification
2. Risk Management.

Potential hazards could be, but not limited to: contact with dangerous voltages; radiation dosage; soldering fumes; machinery; dangerous chemicals, etc. Where applicable you should investigate and state the legal exposure limits or other legal requirements when dealing with any identified hazards.

If such a document is needed it is typically only two short paragraphs on one page signed by both the student and supervisor

2.3 – Time management

You are expected to show a high level of engagement with and commitment to your project. Success requires self-motivation and a professional attitude to the work. During daytime periods for which no formal lectures, laboratories nor tutorials have been scheduled, you are expected to be occupied on your project and it is expected that you, in so much as is possible, work in the project space provided so that you can be easily found by your supervisor or other staff members for non-scheduled meetings/discussions. You are reminded that you should carefully balance the time spent working on your project against the time spent studying for your other modules.

2.4 – Keeping a Log-Book

As part of the project, you are required to keep a dated logbook as a hardback notebook or ring binder. This can be supplemented by an electronic blog if you wish. The log-book is not a formal "lab-report" but acts more like a diary, keeping a basic chronological record of each day's work on the project. You might include notes from background reading, derivations of equations from papers, analysis you have attempted, outlines of diagrams and sketches, flow diagrams of code etc. Any results obtained should be recorded in the log-book, including both positive and negative results. Aside from being good professional practice, you will find the log-book invaluable when you come to write your final thesis, or to give a presentation, as the raw material will be readily available to you.

After each meeting, you will enter some relevant notes summarising the discussion and documentation any decisions made during the meeting.

You may be asked to present the log book for inspection at the oral examination conducted at the end of your project.

2.5 – The role of the project supervisor

The supervisor's role is to advise and monitor your progress. It is not their role to give you precise instructions on what to do next although normally they will indicate promising approaches or recommend against undertaking unproductive or wasteful approaches. If however explicit direction is given then that should be followed. Do not be afraid to come up with your own ideas or suggestions, but make sure to discuss them with your supervisor before committing any significant time or resources to them. It is your duty and responsibility to maintain regular contact with your supervisor.

2.6 – Personal laptop / computer

All students are expected to have access to their own, working, personal laptop / computer, which must be in good condition. Depending on the project type, it may be necessary to install specific software on your computer during the project. You should discuss with your supervisor any special requirements this software might have (e.g. some software may work on Windows, but not on Mac OS). It is your responsibility to ensure this conversation takes place in good time, before the start of your project. The break between Semester 1 and Semester 2 is the ideal time to contact your supervisor for clarification on these issues.

2.7 – Project work space

Project work space is provided where each student has a desk and sufficient space to conduct their studies; it is important to treat this area as a work environment respecting the needs of your fellow classmates. The playing of games or music aloud or any activity that may disturb others is strictly not permitted. Disciplinary action will be taken against any student found disobeying these rules.

2.8 – Equipment and components

Any equipment or components purchased by BDIC for the purposes of project work must be kept in good working condition and treated with respect by all students. Any malfunction or unexpected / unusual behaviour of such equipment must be communicated to your supervisor immediately. Any such equipment will remain property of BDIC.

2.9 – Sources of information

Whatever type of project you are undertaking, you will need to research the background of the subject area. However, before carrying out a literature search there are a number of issues that need to be considered. Relevant information and specific guidelines about starting a project can be found from the UCD Library webpage at:

http://libguides.ucd.ie/infobasics

In order to start a project, up-to-date information on the subject area is normally required. Textbooks may give excellent background information, but they will not necessarily provide detailed information about current developments. Journals will be an important source of information for the project, as they contain details of recent developments in a subject.

All students have electronic access to the most important databases of engineering journals via the UCD Library at: www.ucd.ie/library

It should be noted that the Internet is not always an accurate source of information. Any information gathered in this manner should be critically analysed before being applied in a particular project.

In addition, remember that material contained in your Stage 3 and Stage 4 lecture notes may be relevant to your project work.

2.10 – Insufficient progress

If at any stage your progress is unsatisfactory you may be given a written warning by your supervisor that you are in danger of failing the project module unless you improve your work practices. This is a serious matter, only to be enacted in cases where there is a guanine risk of failure and should not be taken lightly but either the student of staff member. Such warnings are brought to the attention of the Vice-Principal for Teaching and Learning.

2.11 – Problems and seeking help

If you have a problem concerning your project, you should initially approach your project supervisor. In exceptional circumstances you may contact the Vice-Principal for Teaching and Learning. You must not approach another member of staff, postdoctoral fellow or postgraduate student for advice or direction on your project without the informing your supervisor in advance.

3. Final Group Thesis

At the end of the semester each project group must submit a joint group thesis detailing all the work completed during the project.

- The format is single-spaced A4 and should be presented as a stapled document.
- A font size of 11pt should be used.
- The margin size should not exceed 2.54cm – the default in Microsoft Word.
- The presentation of your reports is important, with the correct use of paragraphs and headings; the correct positioning of figures; the formatting of text; the use of capitals, bold, italics etc.; and the inclusion of references and use of the correct referencing format.
- Diagrams, graphs etc. should be used where appropriate.
- It is expected that the length should be approximately 50 pages depending on group size with at least 10 pages (c. 2500 words) in each student's individual section.

3.1 – Correct use of References

In preparing your reports and presentations it is important to correctly reference the published work of others. If you do utilise AI in the preparation of any of your assignments, you must include a section in your introduction or early in your submission explaining what you used AI for, and the prompts you used to surface your results. Failure to do this will be regarded as a violation of the academic integrity rules already in existence and will be penalised accordingly.

There are various ways of citing references. It is recommended that the IEEE referencing style be used for project reports and presentations. Guidance on the correct use of reference styles can be found at:

IEEE Guidelines: https://ieee-dataport.org/sites/default/files/analysis/27/IEEE%20Citation%20Guidelines.pdf

General information from UCD Library: http://libguides.ucd.ie/academicintegrity

All references must be correctly numbered in the main body of your reports and presentations e.g. "as discussed by Smith [1]" or "Smith [1], [2]" or "as discussed in [1], [2]", where [1] or [2] refer to the corresponding reference numbers in the main list of references in the report.

3.2 – **Final Report Layout**

As the report is a group submission it will need to comprise of a "shared" section and several "individual" sections each clearly labelled.

- Shared: Introduction (in which the aims of the project are presented).
- Shared: Literature Review (the students' own description of up-to-date developments in the field).
- Shared: Description of work division between the group members. (Depending on the nature of your project, this may also include a high-level design of your project to show how the work of individual group members fits together).

- Multiple Individual Sections: containing:
  - Clear declaration of authorship
  - Description of all the work completed.
  - Results
  - Discussions
  - Conclusions

- Shared: Overall Discussion and Conclusions.
- Shared: Suggestions for possible future work.
- Shared: References using the correct IEEE format.
- Shared and or individual: Appendices (containing e.g. all lengthy mathematical derivations and verbose technical details).

The above should be seen as a guide, and the nature of each project differs and may require a very different structure. Please ask for advice from your supervisor for guidance.

3.3 – **Final report and software submission procedure**

A PDF copy of the report should be submitted by each student (the same copy by each student) before the appointed time (see 0). These must be accompanied by a signed Final Report Receipt (see Appendix A for a copy of the Receipt). The submission will be made to Brightspace.

The PDF files should be searchable so the included text may be interpreted by a computer. By submitting your thesis using this method, it will automatically be checked for plagiarism of any material included in the report.

It is recommended that any software code should be included as a .zip file and uploaded to Brightspace also. The files uploaded should be complete and in the correct format, i.e. it should be possible for the examiner to download, compile (if applicable) and run.

4. Oral Examination

The oral examination will be conducted with students examined individually. The examination will be performed in a room with the student under examination, the project supervisor and thesis examiner present. The examination is open book and students can refer to code, equations, definitions and or statements from their thesis or other sources during their examination where the examiners allow. Students may not read from preprepared statements or notes throughout the exam and must comply with examiners directions on use of material.

5. Presentation

The presentation should contain an introduction to the problem addressed, a brief overview of the background to the problem including, a clear statement of the project aims, an outline of the methods / approach used and details of results.

When preparing your presentation, you should consider your classmates to be your target audience, i.e. your presentation should be accessible to an audience with a general Internet-of-Things/Electronic Engineering background, who are not necessarily expert in the specific field.

The presentation will be assessed on both presentation quality and on technical content; it is important, therefore, that you attend the seminar series on Presentation Skills during which guidance on preparing a presentation will be given.

Some important guidelines for the presentation are:

- Duration should be max. 15 minutes, with **all project members** involved equally in the presentation. It is expected that single person projects may be shortened (e.g. 10 mins).
- For organisational reasons, presentations will be held in multiple sessions. The audience will consist of those classmates (peers) also presenting in that session, other BDIC students, BDIC staff members.
- A judging panel consists of BDIC academic staff who will also be present.
- Diagrams, graphs etc. should be used where appropriate.
- All presentations must be loaded and tested on a designated computer in good time before the start of the session you are scheduled for. Note that students are not permitted to use their own computer for these presentations. Exceptions to this rule are permitted for presenting project demonstrations, however, in such case the responsibility lies with the students to ensure in advance smooth operation – no additional time will be allocated for this purpose.
- Each presentation will be followed by a short question and answer session.
- All Stage 4 students are encouraged and expected to attend all the presentation sessions their timetable permits.

– Key Dates

The following dates indicate important milestones for the project – late submissions are dealt with according to the rules set out in Appendix B.

Item	Submit	Due Date
Final Report.	- PDF submission to Brightspace.	3pm, Friday, week 13.
Project presentation	- Submit presentation slides (or other presentation material) - Deliver presentations	Slides delivered by 5pm, Monday of week 14 Presentations delivered week 14
Oral examinations	- N/A	During week 15.

Appendix A. Project Report Receipt

On the following page, you will find a cover sheet for use with final project thesis.

Please complete and the submit with your thesis.

BE (IoTE) Degree Program

Stage 4 Project Group Thesis Receipt

Project Title:

Supervisor:

Student Name(s)	UCD Student Number	UCD Student Number

Plagiarism: the unacknowledged inclusion of another person's writings or ideas or formally presented work (including essays, examinations, projects, laboratory presentations). The penalties associated with plagiarism designed to impose sanctions seriousness of University's commitment to academic integrity. Ensure that you have read the University's ***Briefing for Students on Academic Integrity and Plagiarism*** and the UCD ***Statement, Plagiarism Policy and Procedures***, (http://www.ucd.ie/registrar/)

Declaration of Authorship

I/we declare that all of the following are true:

1) I/we fully understand the definition of plagiarism.
2) I/we have not plagiarised any part of this project and it is my original work.
3) All material in this report is my/our own work except where there is clear acknowledgement and appropriate reference to the work of others.

Signed………………………………………… Date ……………………………………

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Signed………………………………………… Date ……………………………………

Office Use Only

Date and Time Received:

N/A

Received by: Brighspace Submission

Report Tracking

Appendix B. Late Submission Policy

Submission deadlines are important for many reasons. Planning and organising work to have it completed by a deadline is an important skill and important for a professional's development. It is important on grounds of equity, in that it is unfair for students to gain advantage by choosing to submit their work late. Deadlines are also necessary for the efficient running of the system. Notwithstanding UCD's exceptional circumstances policy (application to be made through the BDIC academic affairs office) the following deadlines will be strictly enforced:

<u>Final Report</u>

To allow sufficient reading time for the accessors it is necessary to have a very strict late submission policy for the project final report:

1. If the report is submitted at any time up to 24 hrs after the due date (i.e. up to 3pm on the Saturday directly following the due date), the grade awarded will be reduced by one grade point (for example, from B- to C+) ;
2. If the report is submitted more than 24 hrs but up to 48 hrs after the due date (i.e. up to 3pm on the Sunday directly following the due date), the grade awarded will be reduced by two grade points (for example from B- to C);
3. Reports submitted more than two days late will not be accepted and will attract an NG (no grade) for the entire group.

<u>Presentation</u>

To allow for presentation preparation time for the accessors it is necessary to have a very strict late submission policy for the presentation slides:

4. If the slides (or presentation material) are submitted at any time up to 24 hrs after the due date, the grade awarded will be reduced by one grade point (for example, from B- to C+) ;
5. If the slides are submitted more than 24 hrs but up to 48 hrs after the due date, the grade awarded will be reduced by two grade points (for example from B- to C);
6. Slides submitted more than two days late will not be accepted and will attract an NG (no grade) for the entire group.

Appendix A – Plagiarism

A substantial proportion of academic work generally involves making use of the published and unpublished work of others. This is an everyday part of academic and practical/industrial work. The creation of knowledge and wider understanding in all academic disciplines depends on building from existing sources of knowledge.

In the preparation of your academic work, where the ideas or words of other individuals are being used, you must include a clear acknowledgement of the source of information (see above reference example).

On the other hand, plagiarism is the inclusion of another person's writings or ideas or works, in any formally presented work (including essays, theses, project reports, laboratory reports, examinations, oral, poster or slide presentations) without due acknowledgement, either wholly or in part, of the original source of the material through an appropriate citation. Even the unintended misuse of material without formal and proper acknowledgement can constitute plagiarism.

For information on how to avoid plagiarism and the UCD policy and procedures on plagiarism visit:

http://libguides.ucd.ie/academicintegrity/

https://www.ucd.ie/t4cms/Plagiarism_Policy_Academic_Policy_2005.pdf

UCD acknowledges that plagiarism is a very serious academic issue and is subject to disciplinary actions and penalty in grades. You are therefore strongly advised to review all of your reports and presentations for instances of plagiarism before submitting them.

If plagiarism is detected in any submitted written aspect of your project, appropriate action as described in the UCD Pplagiarism Policy and Procedures document (see above link) will be taken.

Appendix B – Indicative Grading Scheme

Grade	Criteria
A+ A A-	**A+** : Exceptional project and report, with aspects potentially publishable in international refereed journals. Excellent in all major aspects such as personal commitment, scale of achievement, quality of report (including review of state-of-the-art) and level of comprehension demonstrated at interview. Hardware aspects containing original or innovative ideas and/or marked by high quality, professional-level implementation and presentation. Software aspects involving original code embodying new ideas, algorithms or analytical concepts. Theoretical/simulation predictions independently verified e.g. by experiment. **A-/A**: Very high quality project and report. Significant independent extension or development beyond reasonable initial project goals, taking account of nature of project and starting baseline of student knowledge. Evidence of high level of success in problem-solving. Challenging hardware design or system implementation achievements. Software achievements beyond routine application of commercial tools/packages, such as developing substantial working original code, professional-level user interface etc. In the A-grade range a high quality professional-level report will be expected, well-organised and clearly written (including figures, spelling etc), with a clear presentation and discussion of results,
B+ B B-	**B+:** High-quality project by a student showing clear potential as a successful research postgraduate student or as a highly competent professional engineer. Above-average achievements in terms of involvement, problem-solving and work completed, taking account of nature of project and starting baseline of student knowledge. High quality report and sound understanding of subject matter of project. **B-/B**: Strong project achieving most major project goals with clear evidence of achievements that could be reasonably be expected in terms of involvement, problem-solving and work completed, taking account of nature of project and starting baseline of student knowledge. Well-organised report demonstrating a very good general knowledge and understanding of the subject matter.
C+ C C-	Satisfactory/ good project. Competent report demonstrating adequate but not complete knowledge and understanding of the subject matter. Work assigned competently performed with serious effort to overcome obstacles and achieve substantive results. Commendable achievement but with some identifiable weakness in one or more major aspects of the project (e.g. insufficient commitment or engagement with project, unsatisfactory report quality, significant failures of understanding evident at interview). Adequate plan and reasonable capacity to apply knowledge to problem solution but with some errors. Some critical awareness and analytical qualities. Generally good presentation with some presentation errors.
D+ D D-	Project showing evidence of work done and acceptable levels of achievement, but with several serious weaknesses in important aspects of the project. Acceptable report demonstrating some basic knowledge and understanding of the subject matter with acceptable level of presentation errors. Limited use of evidence or citation and little critical awareness displayed.
E+ E E-	Marginal project. Fails to meet minimum acceptable standards yet engages with subject matter, albeit with major deficiencies. Insufficient evidence of adequate body of work performed or project showing unacceptable levels of commitment, understanding or achievement. Inadequate presentation with many presentation errors.
F+ F F-	Unacceptable project. Failure to address the problem. Some knowledge displayed but many omissions, errors and major inaccuracies. Unacceptable standard of presentation.
G+ G G-	Wholly unacceptable. Complete failure to address problem. Little or no knowledge displayed. Completely unacceptable standard of presentation.

NG	No grade